

Emma Vellard

Planetary Scientist & Space Systems Engineer

Los Angeles, Unites States

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Feel free to visit my portfolio:

emmavellard.com

EDUCATION

TAS Astro Advanced Master

ISAE Supaero | 2025 - 2026

Space Studies Program (SSP)

International Space University | 2025

MSc Planetary Sciences and Space Exploration

Paris Saclay University | 2022 - 2023

MSc Space Science and Technology

Paris Observatory | 2021 - 2022

Bachelor's degree Fundamental Physics

Clermont Auvergne University | 2019 - 2021

Preparatory class for engineering school (CPGE)

Blaise Pascal Highschool | 2018 - 2019

Scientific degree, Engineering Science

Louis Rascol Highschool | 2015 - 2018

PROJECTS

Lunar Infrastructure for Sustainable

Exploration Study of technologies and architectures enabling sustainable human presence on the Moon, including ISRU, logistics, and exploration systems.

SolarConflux Python tool for identifying coordinated observation opportunities across multiple heliophysics missions.

CERTIFICATES

Machine Learning Specialization

Stanford University | 2024

Mathematics for Machine Learning

Imperial College of London | 2024

LANGUAGES

French (Native)

English (C1 - 98 TOEFL iBT)

Japanese (Beginner)

Spanish (Intermediate)

HOBBIES AND INTERESTS

Volleyball, Bouldering, Karate

Photography

Scuba Diving - PADI Open Water Diver

Planetary scientist and space systems engineer with experience at NASA JPL, ESA, and Tohoku University. Specialized in planetary interior modeling, mission science, scientific software development, and space mission design, with a strong interest in future lunar and planetary exploration.



EXPERIENCE

2026 | NASA Jet Propulsion Laboratory (NASA JPL)

• Visiting Student Researcher | United States (6 months)

- Develop geochemical and planetary interior models of the icy moons to investigate subsurface oceans, internal structure, and habitability. Contribute to the open-source PlanetProfile framework while advancing approaches applicable to future lunar and planetary exploration.

2022-2025 | European Space Agency (ESA)

• Young Graduate Trainee: Juice & BepiColombo missions | Netherlands (2 years)

- **Juice:** Developed a centralized scientific data and knowledge platform to support mission preparation activities for ESA's JUICE mission and streamline access to mission-related information.
- **BepiColombo:** Developed SolarConflux, a Python tool for identifying coordinated solar observation opportunities across multiple heliophysics missions.

• Embedded Software Engineer: YPSat ESA mission | Netherlands (1 year)

- Part of the software development team of the Wake-Up Subsystem which is tasked with autonomously identifying the initiation of the Ariane 6 launch sequence and initiating the remainder of the YPSat system.

• Intern: BepiColombo ESA mission | Spain (5 months)

- Prepare upcoming Mercury observations using the MeSS database based on MESSENGER mission data and perform data processing to support BepiColombo science operations.

2023 | Tohoku University

• Planetary Science Research Intern: Phobos composition analysis for the MMX JAXA mission | Japan (4 months)

- Investigate the origin hypothesis of Phobos and the Martian moons through spectral analysis and analogue rock sample studies to support the MMX mission's compositional objectives.

2022 | Atmospheres and Space Observations Laboratory (LATMOS)

• Research Scientist: Design of a space mission to study Ceres | France (1 year)

- Study of the geomorphology of Ceres and the cryovolcanism at the surface.
- Dimensioning of a sounder radar in a 3U cubsat.

SKILLS

Scientific Expertise

Planetary Science, Planetary Interior Modeling, Planetary Geophysics, Planetary Surfaces, Geochemistry, Spectroscopy, Mission Science, Lunar Exploration

Technical Skills

Python, C, Scientific Software Development, Numerical Modeling, Data Analysis, Pandas, Machine Learning

Space Systems Engineering

Space Mission Design, Space Systems Engineering, Embedded Software, Human Spaceflight, Knowledge Management, Project Management

Communication skills

Outreach communication, Oral and written skills to present and synthesize scientific work